

Diagnosing competitive reactions using (aggregated) scanner data *

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We study competitive response functions with scanner data on price and promotional activities. Causality tests are used prior to parameter estimation. The results indicate that price and feature have statistically significant causal effects more frequently than other promotional variables and they have disproportionately greater frequencies for quick reactions relative to other instruments. Multiple competitive reactions are common, although the same marketing instrument is more likely to be used to react than a different one. Reactions occur with decreasing probability over time suggesting distinctions between retailer- and manufacturer-dominated reactions. The results can also be used to identify other strategic differences, such as the classification of competitors as leaders and followers.

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Introduction

Marketing is by its very nature concerned with the interaction between firms and the marketplace. During the last decade developments in strategic marketing, planning and decision making have received considerable attention. The emerging literature on strategic market planning (Abell and Hammond, 1979), marketing strategy (Wind and Robertson, 1983), marketing theory with a strategic orientation (Day and Wensley, 1983) and strategic market management (Aaker, 1984) enriches and broadens the marketing discipline. Also, the inclusion of the term "market" into phrases such as strategic market planning and strategic market management serves to emphasize that strategy considerations need to be driven by the market and its environment (and only secondarily by factors internal to the firm). In addition, the process should be *pro-active*, in the sense that management attempts to shape its environment (e.g. Aaker, 1984, p. 13), as well as reactive.

In this period of ferment, new theories, new paradigms, new concepts and new tools are being developed. Research agendas proposed by Day and Wensley (1988) and Wind and Robertson (1983) include topics such as competitive market behavior, the development of offensive and defensive marketing strategies, and the formulation and application of methods to assess competitive actions and reactions. Thus the emphasis is on a marketplace that includes customers *and* competitors. The growing attention to competitors can be explained by the decline in many Western economies in the early eighties. Before this decline firms could take ad-

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vantage of the fact that most markets were growing. In the "flat eighties" the realization that sales gains would often come only by taking share away from competitors (Kotler, 1988, p. 234) resulted in greater attention to competitors' actions and reactions.

There is also increasing interest in understanding competitive actions and reactions for policy and antitrust decisions. For example, in the airline industry firms appear to be especially interested in maintaining dominance in territories where they have a large market share. US airlines have been accused of signaling their intent to follow an aggressive pricing strategy in an area dominated by a competitor if that competitor introduces lower fares in an area dominated by the firm. Eastern, the only airline responding in public to the accusation, claimed that the firm sets its prices independently of other airlines. One might argue, however, that independence should occur only under monopoly conditions. In any event, knowledge about competitive actions and reactions can be used to understand the nature of competition in an industry.

Notwithstanding the shift in attention from customers to customers *and* competitors, Wind and Robertson (1983) maintain that systematic treatment of competitive analysis is lacking (see also Day and Wensley, 1983). In the econometric literature in marketing, models have been used primarily to quantify customer reactions to (changes in) marketing decision variables. Due to the availability of scanner data, more elaborate and detailed models are applied. And while managers are increasingly interested in using the results from econometric models, little is included on the measurement of competitive reactions (e.g. Abraham and Lodish, 1987).

Pro-active marketing strategies can be performed and implemented only when (brand) managers know both how customers and competitors may react. Using only the estimated own effects to determine appropriate

marketing activities for a given brand would be incorrect if competitors employ their own activities in response to actions undertaken. Thus, it is necessary to estimate how competitors react to each other's marketing programs (see also Hanssens, 1980). Rarely, however, is information on competitive reactions provided along with estimated sales or market share effects. One reason for this is the lack of sufficient data. For example, aggregation of data over time hides some of the variability in marketing activities required to estimate competitive reactions. Another reason offered is the difficulty of accommodating "...many different types of competitive actions..." (Carpenter et al., 1988). However, with weekly scanning-based store data aggregation over time is not a concern, because prices and other marketing variables usually do not vary within a week for a given brand.

Objective of the study

The objective of our study is to model competitive reactions using weekly scanner data. The face validity of the results can be ascertained by the extent to which parameter estimate signs agree with prior expectations. In addition, we specify hypotheses about the specific marketing variables involved. These hypotheses have been formulated, in part, based on discussions with industry experts (i.e. product managers and market researchers).

In the study reported here we analyze competitive reactions using weekly scanner data collected at the retail level. In Europe the growth in retailer concentration in many types of distribution outlets has accelerated the shift in control of certain marketing instruments from manufacturers to retailers (see Nooteboom et al., 1986). Price decisions for a brand, for example, are made ultimately by retailers, not manufacturers. And

the occurrence of temporary price cuts, premiums and displays critically depends on the degree of acceptance (pass through) of promotional programs by retailers. Thus, under these conditions the modeling of competitive reactions has to reflect the role of manufacturers and retailers.

Competitive reactions can be represented in reaction matrices, showing the effects of decisions made for brand i on decisions made for brand j , $i, j = 1, \dots, n$, $j \neq i$. For example, the manager of brand i should know how its $(n - 1)$ competitors react to a marketing program for i . Assuming k decision variables for each brand, $k(n - 1)$ reaction relations are required before the manager can evaluate "what if" scenarios involving alternative marketing programs.

In the next section we review previous studies of competitive reactions based on reaction matrices. In the third section we describe our model, the hypotheses and the empirical approach. This is followed by the results and a discussion of the limitations of our study in the fourth section.

Competitive reactions

The use of reaction matrices to diagnose competitive reactions was pioneered by Lambin, Naert and Bultez (1975). In their (LNB) model, a distinction is made between simple and multiple competitive reactions. A *simple* competitive reaction refers to a case where a competitor reacts with the same instrument as the one used by the firm inducing the reaction. For example, a change in the price for brand i results in a change in the price for brand j . A *multiple* competitive reaction is one where a competitor may react to a change in the price for brand i not just by a change in price for j , but also by changing

the advertising budget, the advertising message, product quality, promotions, etc.

The very nature of reactions implies there is a time lag in modeling reaction functions. However, to some extent competitors' moves may be anticipated, with the possibility that reactions occur before or at the same time the relevant actions are registered.¹ And, dependent upon the data interval available for empirical research, some reaction functions may appear to be instantaneous. In general, however, fast reactions are more likely for marketing variables such as price than for variables such as manufacturers' advertising, samples, and refunds that require lead time.

Another critical aspect is the set of assumptions about competitors. Competitive reactions can be determined based on different models.² In the "classical" LNB model, it is assumed that there is one market leader, the other competitors being followers. Furthermore, these competitors are treated as one group, so the market is defined as consisting of two parties: the leader firm and the aggregate of the other firms. Strictly speaking, the desirability of marketing actions is easily determined for the leader, in this case. For example, assuming only one decision variable (price), two brands, instantaneous reactions, and no lagged effects, the brand demand elasticity with respect to own price is equal to the own price elasticity plus the cross-price elasticity multiplied by the reaction elasticity (the competitor brand's price with respect to own brand's price).

¹ Anticipation of competitors' moves may not occur frequently in a mature product category with fairly stable market shares.

² See for other models than the ones which are discussed in this section: Wildt (1974); Gatignon (1984); Plat (1988); Plat and Leeflang (1988). For a recent review of competitive reactions see Hanssens et al. (1990, p. 201-210). Also, Robinson (1988) and Gatignon et al. (1989) developed models to estimate marketing mix reactions to new entrants.

In the LNB model, multiple competitive reactions are estimated by calibrating the following relations:³

for $l = 1, \dots, k$:

$$U_{lt} = f([u_{hit}]_{h=1, \dots, k}), \quad t = 1, \dots, T, \quad (1)$$

where

U_{lt} = aggregate or average competitive value of the l th marketing instrument in period t ,

u_{hit} = value for the h th marketing instrument of the leader in period t .

For illustrative purposes, we define elements of equation (1). For example, let

$$U_{1t} = \frac{\sum_{j=2}^n P_{jt}}{n-1} \quad \text{and} \quad u_{1t} = P_{1t}, \quad (2a)$$

and

$$U_{2t} = \sum_{j=2}^n A_{jt} \quad \text{and} \quad u_{2t} = A_{1t}, \quad (2b)$$

where

P_{jt} = price of brand j in period t ,

A_{jt} = advertising expenditures for brand j in period t , $j = 1, \dots, n$,

n = total number of brands.

If price and advertising are the only relevant marketing instruments, two equations, each with two predictors, are required. Also, it is easy to see that a model with simple competitive reactions is a special case of equation (1).

In a number of extended LNB models, no distinction is made between leaders and followers. Importantly, all brands can be considered separately in what amounts to a decomposition of competitive interactions.

Leeflang and Reuyl (1985) determined the interactions for each set of two brands:

for $l = 1, \dots, k$,

$i, j = 1, \dots, n$; $i \neq j$:

$$u_{ljt} = g([u_{hit}]_{h=1, \dots, k}), \quad t = 1, \dots, T, \quad (3)$$

where

u_{ljt} = value of the l th marketing instrument for brand j in period t .

In this extended model, the number of equations to be calibrated equals $kn(n-1)$, compared with k equations for the LNB model. A difficulty with this model is that although all brands are allowed to compete with each other, each of the equations estimated uses predictors for only one (competing) brand. Thus, the parameter estimates in (3) may be biased due to omitted predictor variables (see for an example Wonnacott and Wonnacott, 1981, p. 99). Also, the system of equations may be underidentified. This can happen if for a given pair of variables both variables serve as criterion and predictor variables.

An alternative extension used by Hanssens (1980) is:

for $l = 1, \dots, k$ and $j = 1, \dots, n$:

$$u_{ljt} = g\left([u_{hit}]_{\substack{h=1, \dots, k \\ i=1, \dots, n}} - [u_{ljt}]\right), \quad t = 1, \dots, T. \quad (4)$$

For equation (4) the number of relations to be estimated is kn . While there are fewer equations to be estimated in (4) compared to (3), multicollinearity and degrees of freedom may cause a problem in estimating (4). For example each equation may have $(kn-1)$ predictors, without considering time lags. Thus, $(kn-1)$ has to exceed T . And, if competing brands react similarly, those variables may show a high degree of correlation. Equation (4) does allow for joint marketing decision making when $j = i$, i.e., the possibility that changes in one variable are accompa-

³ In the relations (1)–(4) it is assumed that all variables relate to the same period. It is, of course, also possible that reactions occur after a time lag.

nied by changes in one or more *other* variables for a given brand.

We consider the problem of incompletely specified equations for equation (3) to be more serious than the multicollinearity or other potential problems for (4). Thus, we want to examine competitive reactions using a model that is as completely specified as possible. This does mean, however, that some preliminary data analysis is required to reduce the number of potential predictor variables to a manageable set. To illustrate the problem, with seven brands and five marketing variables, assuming all reactions to be instantaneous, there are 34 predictor variables. By allowing also for lags in the reactions, the number of predictors quickly exceeds typical sample sizes. To overcome this problem, and to strengthen the validity of empirical results, we use causality tests for all pairs of variables (see below). The causality test results are the basis for the competitive response function specifications.

Empirical analysis

By calibrating competitive reactions at the *retail level*, we describe the nature of competition between seven large brands ($n = 7$) of a frequently purchased nondurable, nonfood, consumer product sold in the Netherlands. The product is a necessity, and the seven brands together account for approximately seventy percent of the market. The brands are available in different varieties and package sizes in many retail outlets. For the empirical analysis we use a relatively homogeneous subset of all alternatives. For example, all unusual varieties and package sizes are excluded. We assume that these varieties and package sizes are irrelevant to the estimated competitive reactions. This assumption is supported by managers responsible for brands in this product category. Only market-level data (i.e. data aggregated across

retail outlets) are available for each item (e.g. brand, variety and package size combination) on a weekly basis. To facilitate the empirical analysis, we define variables at the brand level. Aggregation over varieties and package sizes is accomplished by weighting the data by sales at the variety-package size level.

Competition between brands in the product category is dominated by price and promotional programs. Hence it is appropriate to specify competitive reaction models to explain fluctuations in (retail) prices and promotions. We specify the reaction functions in terms of multiple competitive reactions. The competitive actions and reactions are expressed in "changes". For example, for price we use the logarithm of the ratio of prices in two successive periods. This is based on the idea that price changes for competitors having different *average* price levels are more comparable on a percentage than an absolute basis. Promotional activities are specified in terms of simple differences, because of the occurrence of zero values. By using first differences or ratios, possible multicollinearity due to common trends in the variables is avoided. Also, this representation is consistent with the idea that reactions occur in the form of changes from previous conditions. On the other hand, the error term assumptions change at the same time. These assumptions are tested through the application of common test statistics.

Price changes are assumed to occur because of a deliberate strategy set by brand and/or store management. These changes may occur in response to changes in the marketing mix variables of competing brands. Changes at the retail level may be encouraged (as in trade promotions) by the manufacturer. It would be desirable to distinguish between temporary price changes (e.g. due to promotions such as temporary cents-off) and permanent price changes. The data, however, make this impossible. Thus, the

price of brand j in period t can be thought of as a weighted average of the promotional and regular prices, with the weights representing unit sales for the two prices. However, most of the price movements occur as a result of promotions. Implicitly the average price also reflects the degree to which retailers accept trade promotions when these occur. The lower the degree of acceptance the higher the average retail price, *ceteris paribus*.

Hypotheses

We describe hypothesized competitive reactions primarily for price variables. Generally speaking, we expect the same type of reaction function if a promotional variable is substituted for one of the price variables, except for a change in the sign (and magnitude) of the parameter. For example, if a price *decrease* for brand i is hypothesized to cause a price *decrease* for brand j (perhaps after a lag) in a simple competitive reaction, we may also hypothesize it to cause a promotion *increase* for brand j as part of multiple competitive reactions.

We consider three kinds of *price reactions*.

1. Parallel price movements

These represent price fluctuations of brand i , $i = 1, \dots, n$, which develop parallel to the price fluctuations or promotional expenditures of other brands $j = 1, \dots, n$, $j \neq i$. Such parallel movements may occur because of planned price movements, and may also reflect expectations about competitors' brands. For example, some retail chains may offer a price promotion for brand i at the time other chains do this for brand j . Such movements often result from an annual plan developed and agreed upon by management of a brand and the retail chains. Or, the manufacturer of different brands may coordinate *part* of the brands' marketing programs, resulting in parallel movements. We assume that parallel movements refer to the same time period t

or occur with a maximum delayed response of one time period. Parallel price movements may result in a positive relation for competing brands' price variables and a negative relation between one brand's price movements and another brand's promotional expenditure changes.

2. Retailer-dominated price reactions

In case of frequent promotional activities in a product category, it is possible for a retailer to run a promotional activity for one brand in one week followed by an activity for another brand in the next week. The result is that price decreases for brand i occur together with price increases for brand j , $j \neq i$ (or decreases in promotional expenditures for j). In this case we expect a negative relation between price movements for competing brands and a positive sign for one brand's price movements related to another brand's promotional expenditure changes. We assume that such reactions occur in the *short run*, either simultaneously or with a maximum lag of one period.

In the *longer run*, within 2–4 weeks, retailers may react with price and/or promotion movements for brand i to similar movements for competing brands. If these activities are motivated by manufacturers' trade promotions, the nature and frequency of such activities for competing brands indirectly incorporate competitive reactions for manufacturers. In this case we expect partial relations with the same signs as those for parallel movements.

3. Manufacturer-dominated price reactions

In their extreme form, when manufacturers determine prices, these reflect the classical reactions assumed in the models described in the section on competitive reactions. With scanner data (measuring *retail* sales and retailers' activities), these reactions can occur only in case of cooperation between retailers and manufacturers. The reactions may be the result of an adaptive plan-

ning procedure. This means that planning is undertaken to produce changes in the underlying relations between manufacturers and retailers to create a more desirable future (see, for example, Jain, 1985, p. 11). Adapting an annual plan takes, we assume, 5–10 weeks. We expect again partial relations with the same signs as those for parallel movements.

We show in Table 1 a summary of the expected signs and the delays proposed for the categories of multiple competitive reactions. In the empirical analysis we discuss tests for the nature of delayed responses.

We now turn to a description of the *non-price promotional variables* (and their possible competitive reactions). Data on the non-price promotional variables are obtained using a double weighting scheme. The variables are not only weighted over varieties and package sizes but they also reflect weighted distribution figures which refer to a particular promotional activity. The following kinds of promotional activities are distinguished:

ws_{it} = weighted sampling (premium, multi-pack) for brand i , in week t ,

wr_{it} = weighted refund activity for brand i , in week t ,

wb_{it} = weighted (package) bonus offer for brand i , in week t ,

wf_{it} = weighted feature (retailer advertising) activity for brand i , in week t .

The variable ws_{it} is defined as follows:

$$ws_{it} = \left(\sum_{v=1}^{V_i} \sum_{q=1}^{Q_{vi}} S_{qvrit} d_{qvrit}^s \right) / \left(\sum_{v=1}^{V_i} \sum_{q=1}^{Q_{vi}} S_{qvrit} \right), \quad (5)$$

where

S_{qvrit} = unit sales for package size q , variety v , brand i , in week t ,

d_{qvrit}^s = weighted distribution of the sampling activity for package size q , variety v , brand i , in week t , i.e. the percentage of retail outlets offering the sample activity, with each

Table 1

Signs and indications of delayed response of price (p_{it}), and non-price promotions (ws_{it} , wr_{it} , wb_{it} and wf_{it}) for the different kinds of competitive reactions

Criterion variable brand i	Reaction	Source brand j ($j \neq i$)	Sign/period ^a										
			t	$t-1$	$t-2$	$t-3$	$t-4$	$t-5$	$t-6$	$t-7$	$t-8$	$t-9$	$t-10$
price	parallel	price	+	+									
		non-price promotion	-	-									
price	retailer-dominated short-run	price	-	-									
		non-price promotion	+	+									
price	retailer-dominated long-run	price			+	+	+						
		non-price promotion			-	-	-						
price	manufacturer-dominated	price						+	+	+	+	+	+
		non-price promotion						-	-	-	-	-	-
non-price promotion	parallel	price	-	-									
		non-price promotion	+	+									
non-price promotion	retailer-dominated short-run	price	+	+									
		non-price promotion	-	-									
non-price promotion	retailer-dominated long-run	price			-	-	-						
		non-price promotion			+	+	+						
non-price promotion	manufacturer-dominated	price						-	-	-	-	-	-
		non-price promotion						+	+	+	+	+	+

^a Empty boxes indicate we assume that these reactions do not occur.

outlet weighted by its product category volume,

V_i = total number of varieties of brand i ,

Q_{vi} = total number of package sizes of variety v of brand i .

The variable wr_{it} is defined in a similar manner as ws_{it} . The definition of wb_{it} is somewhat more complicated because the percentage of retail outlets offering the bonus package with each outlet weighted by its product category volume is also multiplied by the percentage of regular size added to the package. The retailer promotion variable wf_{it} is defined without the package size index (q), because featuring is not measured at the package size level. This variable is weighted by the annual turnover of the product category in the retail chains featuring brand i in period t .

We propose similar reaction patterns for the promotional variables as those formulated for price. Hence we have: (a) parallel (non-price promotional) reactions; (b) short- and long-run retailer-dominated reactions; (c) manufacturer-dominated reactions.

The simple and multiple competitive reactions of the non-price promotions have the same sign as the corresponding price reactions discussed before (see Table 1). That is, if a competitor's price is expected to have a positive effect on own brand's price, then the competitor's non-price promotion is expected to have a positive effect on own brand's non-price promotion.

We note that we show in Table 1 only the signs of effects for the different kinds of competitive reactions, with the same signs for the different types of non-price promotions. The pattern is similar for price, except that the sign is the opposite of that for non-price promotions (for the same criterion variable).

If the product category could be identified, we would be able to specify hypotheses with regard to, for example, brands that

should emerge as leaders. For example, a leader may be defined as a brand that has many significant effects for its predictor variables in other brands' equations. Due to confidentiality considerations, we cannot do this. Thus, we limit ourselves to an interpretation of the empirical evidence with respect to possible leaders and followers in this industry.

Of the five variables, we expect that price and feature are the most frequently used variables for competitive reactions (H1). This is, in part, because retailers can manipulate these variables easily without manufacturer input. We also hypothesize that competitors are more likely to react with the same than with a different marketing instrument (H2). This is based on the idea that it is natural for managers to use instinctively the same instrument a competitor does. In addition, managers may believe that activities with other instruments do not receive the same attention.

We expect that reactions, if they occur, have a decreasing probability of occurring as the time lag increases (H3). Suppose firm A takes a marketing action. If this action is believed to affect B's business it is important for B to react quickly for two reasons. One, the faster the reaction the smaller the impact of A's action on the market. Two, the faster the reaction the more clearly B is on record as intending to protect its position. We also hypothesize that the variable used for a response is related to the lag with which the response occurs (H4). Price and feature are expected to have disproportionately greater frequencies for quick reactions relative to the other three variables. This is because price and feature can be manipulated quickly by retailers with or without manufacturer input.

Data

Aggregated market level data are used to calibrate competitive reactions. The data are

weekly data covering a period of 76 weeks. The market research company A.C. Nielsen (The Netherlands) B.V. collects these data in the Netherlands using a sample of about 150 stores. The data indicate that for some brands some promotional activities are not performed. For example, a brand may not have sample products or may not have a refund activity in the time period covered by the data.

Model

The following competitive reaction functions are used:

for $i = 1, \dots, n$:

$$\ln\left(\frac{p_{it}}{p_{i,t-1}}\right) = \alpha_i + \sum_{\substack{j=1 \\ i \neq j}}^n \sum_{t^*=1}^{T^*+1} \beta_{ijt} \ln\left(\frac{p_{j,t-t^*+1}}{p_{j,t-t^*}}\right) + \sum_{t^*=2}^{T^*+1} \beta_{iit} \ln\left(\frac{p_{i,t-t^*+1}}{p_{i,t-t^*}}\right) + \sum_{j=1}^n \sum_{t^*=1}^{T^*+1} \sum_{x=1}^4 \tau_{xijt} (wx_{j,t-t^*+1} - wx_{j,t-t^*}) + \epsilon_{it} \quad (6)$$

where $x = 1 = s$, $x = 2 = r$, $x = 3 = b$, $x = 4 = f$, and $t = (T^* + 2), \dots, T$.

Also, for $i = 1, \dots, n$:

$$\begin{aligned} & (wx_{it} - wx_{i,t-1}) \\ &= \delta_i + \sum_{j=1}^n \sum_{t^*=1}^{T^*+1} \mu_{xijt} \ln\left(\frac{p_{j,t-t^*+1}}{p_{j,t-t^*}}\right) \\ &+ \sum_{t^*=2}^{T^*+1} \theta_{xxiit} (wx_{i,t-t^*+1} - wx_{i,t-t^*}) \\ &+ \sum_{t^*=1}^{T^*+1} \sum_{\substack{y=1 \\ y \neq x}}^4 \theta_{xyiit} (wy_{i,t-t^*+1} - wy_{i,t-t^*}) \\ &+ \sum_{\substack{j=1 \\ j \neq i}}^n \sum_{t^*=1}^{T^*+1} \sum_{y=1}^4 \theta_{xyijt} \\ &\times (wy_{j,t-t^*+1} - wy_{j,t-t^*}) + \Phi_{it} \quad (7) \end{aligned}$$

where $x = 1 = s$, $x = 2 = r$, $x = 3 = b$, $x = 4 = f$, and $t = (T^* + 2), \dots, T$; ϵ_{it} and Φ_{it} are random disturbance terms drawn from normal distributions with zero means and variances σ_{1i}^2 and σ_{2i}^2 , $i = 1, \dots, n$, T^* is the maximum number of time lags ($T^* = 10$), and T is the number of observations available ($T = 76$). All other variables have been defined before.

In (6) and (7) lagged endogenous variables have been included. This has been done for several reasons. From discussion with managers and from an inspection of the data it was clear that periods with heavy promotions were frequently followed by periods with relatively low promotional efforts. Although sales of the product are not subject to seasonal fluctuations, promotional programs may be planned according to seasonality or be otherwise somewhat systematically distributed over a time period. This may be the reason why the inclusion of lagged endogenous variables improved the fit of the relations and reduced the degree of autocorrelation of the disturbances.

The hypotheses about the signs and periods, shown in Table 1, have been summarized in terms of the competitive response parameters in equations (6) and (7) and are shown in Table 2. We also show in Table 2 hypotheses about intrafirm effects. However, in general it is difficult to justify specific signs for intrafirm effects.

Causality tests

An inspection of equations (6) and (7) makes it clear that they are underidentified. Our primary interest is in estimating *causal* relations between variables. Causal relations should be distinguished from relations between jointly distributed variables. Thus, we use (bivariate) causality tests to both reduce the set of predictors (only variables showing statistical significance in the causality tests

Table 2

Hypotheses about the signs of the parameters in (6) and (7)

Parameter	Sign	Period ^a	Brand	Promotional effect
β_{ijt}^*	+	$t^* \geq 3$	$i, j = 1, \dots, n; i \neq j$	—
β_{ijt}^*	+, —	$t^* < 3$	$i, j = 1, \dots, n; i \neq j$	—
β_{iit}^* (intrafirm)	—	$t^* = 2$	$i = 1, \dots, n$	—
β_{iit}^* (intrafirm)	+, —	$t^* \geq 3$	$i = 1, \dots, n$	—
τ_{xijt}^*	+, —	$t^* < 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$
τ_{xijt}^*	—	$t^* \geq 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$
τ_{xii}^* (intrafirm)	+, —	all t^*	$i = 1, \dots, n$	$x = 1, 2, 3, 4$
μ_{xijt}^*	—	$t^* \geq 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$
μ_{xijt}^*	+, —	$t^* < 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$
μ_{xii}^* (intrafirm)	+, —	all t^*	$i = 1, \dots, n$	$x = 1, 2, 3, 4$
θ_{xyijt}^*	+	$t^* \geq 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$ $y = 1, 2, 3, 4$
θ_{xyijt}^*	+, —	$t^* < 3$	$i, j = 1, \dots, n; i \neq j$	$x = 1, 2, 3, 4$ $y = 1, 2, 3, 4$
θ_{xyii}^* (intrafirm) for $x \neq y$	+, —	all t^*	$i = 1, \dots, n$	$x, y = 1, 2, 3, 4$

^a For $t^* = 1, \dots, 11$.

are retained) and to strengthen the validity of the estimated reaction functions.

There is an ongoing stream of literature on the topic of causality. Discussions on this topic refer to definitions of causality and non-causality, the operationalization of the causality concept, the description of causality patterns and causality testing (see, for example, Aigner and Zeller, 1988). The concept of causality we use is the well-known Wiener–Granger causality. This concept is defined as follows:

“a variable x is said to cause another variable y , with respect to a given information set containing x and y , if future y -values can be predicted better using past values of x and y than using the past values of y alone”.⁴

Testing the independence versus the causal relationship⁵ between two or more variables in a system of relations is possible using

different tests, such as the Granger–Wald, Granger–Sargent, and Sims tests. (Hanssens et al., 1990, Ch. 5). We used the Haugh–Pierce test to detect causality in bivariate systems through the white noise ARIMA residuals of the time series x_t and y_t . For a detailed discussion of the procedure see Hanssens (1980).

In our study the intrastructures of the times series

$$\ln\left(\frac{p_{it}}{p_{i,t-1}}\right), \quad i = 1, \dots, n;$$

$$(wx_{it} - wx_{i,t-1}), \quad x = 1, \dots, 4, \quad i = 1, \dots, n,$$

have been determined and the time series are filtered by removing the intrastructures. Then white-noise residuals (or innovations) $a(p_i)$ and $a(wx_i)$, $x = 1, \dots, 4$, result. The ARIMA models estimated to obtain white-noise residuals are shown in Table 3.

From Table 3 it is clear that most ARIMA models have a rather simple (MA(1)-) structure, and in many cases the variables have white noise. For some variables in Table 3 no specification is given. The variables are the non-price promotional variables which are

⁴ Granger (1969); for more recent and more elaborated concepts of this kind of causality we refer to Granger (1988). For a discussion about causality in a marketing context, see Hanssens et al. (1990, p. 157–170).

⁵ See Haugh (1976), Pierce (1977) and Hanssens (1980); we closely follow Hanssens (1980).

not used by some competitors. For example, brands 1, 2 and 3 have not used refunds as a promotional tool. The innovations $a(p_i)$,

$a(p_j)$, $a(wx_i)$ and $a(wy_j)$ are cross-correlated at various lags m , $m = 0, 1, 2, \dots, M$, where a lag of one period refers to one week. For

Table 3
ARIMA models of the variables used

Criterion variable	Brand i	Specification of the ARIMA model ^a (right hand side)	Name of the model
$\ln\left(\frac{p_{it}}{p_{i,t-1}}\right)$	1	$a(p_{1t}) - 0.61a(p_{1,t-1})$ (6.89)	MA(1)
	2	$a(p_{2t}) - 0.44a(p_{2,t-1}) - 0.53a(p_{2,t-2})$ (4.45) (5.56)	MA(2)
	3	$a(p_{3t}) - 0.73a(p_{3,t-1})$ (9.12)	MA(1)
	4	$a(p_{4t}) - 0.25a(p_{4,t-1})$ (2.24)	MA(1)
	5	$a(p_{5t}) - 0.61a(p_{5,t-1})$ (6.86)	MA(1)
	6	$a(p_{6t}) - 0.31a(p_{6,t-1})$ (2.78)	MA(1)
	7	$a(p_{7t}) - 0.33a(p_{7,t-1})$ (2.99)	MA(1)
$(ws_{it} - ws_{i,t-1})$	2	$a(ws_{2t}) - 0.87a(ws_{2,t-1})$ (15.27)	MA(1)
	3	$a(ws_{3t}) - 0.68a(ws_{3,t-1})$ (8.12)	MA(1)
	4	$a(ws_{4t})$	white noise
	5	$a(ws_{5t})$	white noise
	6	$a(ws_{6t}) - 0.26a(ws_{6,t-1})$ (2.30)	MA(1)
	7	$a(ws_{7t}) + 1.05a(ws_{7,t-1}) + 0.75a(ws_{7,t-2}) + 0.77a(ws_{7,t-1} - ws_{7,t-2})$ (11.93) (9.22) (8.43)	ARMA(1,2)
$(wr_{it} - wr_{i,t-1})$	4	$a(wr_{4t})$	white noise
	5	$a(wr_{5t})$	white noise
	6	$a(wr_{6t})$	white noise
	7	$a(wr_{7t})$	white noise
$(wb_{it} - wb_{i,t-1})$	2	$a(wb_{2t})$	white noise
	4	$a(wb_{4t})$	white noise
	6	$a(wb_{6t}) - 0.66a(wb_{6,t-1})$ (7.50)	MA(1)
	7	$a(wb_{7t})$	white noise
$(wf_{it} - wf_{i,t-1})$	1	$a(wf_{1t}) - 0.98a(wf_{1,t-1})$ (95.56)	MA(1)
	2	$a(wf_{2t}) - 0.98a(wf_{2,t-1})$ (105.30)	MA(1)
	3	$a(wf_{3t}) - 0.98a(wf_{3,t-1})$ (91.22)	MA(1)
	4	$a(wf_{4t}) - 0.98a(wf_{4,t-1})$ (100.70)	MA(1)
	5	$a(wf_{5t}) - 0.99a(wf_{5,t-1})$ (98.20)	MA(1)
	6	$a(wf_{6t}) - 0.98a(wf_{6,t-1})$ (83.70)	MA(1)
	7	$a(wf_{7t}) - 0.90a(wf_{7,t-1})$ (91.99)	MA(1)

^a Absolute value of the t -statistic in parentheses.

example, the cross-correlations for competing price variables are shown in equation (8):

$$r_{a(p_i), a(p_j)}(m) = \frac{\sum_{t=1}^{T-m} (a(p_{it}) - \bar{a}(p_i))(a(p_{j,t+m}) - \bar{a}(p_j)) / (T-m)}{S_{a(p_i)} S_{a(p_j)}} \quad (8)$$

for $i, j = 1, \dots, n$; $i \neq j$; $m = 0, 1, 2, \dots, M$, where

$\bar{a}(p_i)$ = sample mean of the white-noise residuals of the time series p_{it} ,

$S_{a(p_i)}$ = sample standard deviation of these time series for $i = 1, \dots, n$,

M = maximum lag.

To test for causal relations between the predictor variables and the criterion variables, we use (shown only for the price variables):⁶

$$S_1 = T \sum_{m=0}^M r_{a(p_i), a(p_j)}^2(m) \quad (9)$$

where $m = 0, 1, 2, \dots, M$, and $M = 10$. The test statistics are approximately chi-square distributed with $(M+1)$ degrees of freedom if the cross-correlated variables p_i , p_j are independent. The null hypothesis of series independence is accepted at level α if $S_1 < \chi_{\alpha, M+1}^2$. Otherwise, the conclusion is that the series (p_i, p_j) are causally related to each other. Similar test statistics can be defined between p_i and wx_j , wx_i and p_j , and wx_i and wx_j (for $i, j = 1, \dots, n$, $x = 1, \dots, 4$). The values of the test statistics have been computed for different values of M . The individual cross-correlations of the pairs of time series at various lags can be inspected in this way. For example, we computed $(M+1)n(n-1)$ values for the test statistic S_1 .

It is clear that it is not very appealing to present all values of all test statistics. We show the values of S_1 for $M = 0, 4, 6, 8, 10$

in Table 4 as an illustration. The critical chi-square values for $\alpha = 0.10$ are 2.71 ($M = 0$), 9.24 ($M = 4$); 12.02 ($M = 6$); 14.68 ($M = 8$); and 17.28 ($M = 10$). The results in Table 4 suggest that the price of brand i , $i = 2, \dots, 7$, depends causally on the price of brand 1. Similarly, the price of brand 1 appears to depend causally on the price of brands 2, 4, 5, 6 and 7. The row values of the test statistic are generally greater than the column values for p_1 . This suggests that the prices of all competing brands may depend causally on the price of brand 1, while the price of brand 1 may depend causally on only some of those brands. However, the evidence about causal relations between brands 1 and 4 is limited to the case where $m = 0$ (i.e., instantaneous causality), leaving the direction ambiguous. In general, the results suggest that, as far as price is concerned, brands 1 and 5 may function as leaders in this product category, while brand 7 is a follower. The price variation for brands 3, 4 and 6 appears to be largely independent of the prices for other brands in this product category.

Because Table 4 shows the results for just one out of 25 possible tables, we summarize the findings for all test statistics in Table 5. The letters indicate which predictor variables are found to have causal effects on a given criterion variable. For example, the price of brand 1 causally depends on the price and feature of brand 2, feature of brand 3, etc.

An inspection of Table 5 makes it clear that there is ample evidence suggesting the existence of multiple competitive reactions. Every row includes variables different from the one defined as the criterion variable. In total there are 256 significant causal relations reported in Table 5 including 35 which pertain to relations between different variables for the same brand. Of the remaining 221 competitive reactions, 65 (29 percent) are simple. The observed percentage is significantly different from an expected percentage of 20 percent ($p < 0.01$) if simple

⁶ The sample size is insufficient for using validation samples to strengthen the causal tests.

and other competitive reactions are equally likely. This is consistent with hypothesis H2.

For all variable combinations we show the number of times brand pairs may be causally related in Table 6. The maximum number for each of the cells in this table is 42 (excluding causal relations between different variables for the same brands). We see that the largest

number, 24, pertains to price–price relations. The smallest number, 2, refers to sampling–refund relations. Thus, the results suggest that brands often react with a change in price when price changes are implemented for a competing brand. And it is very unlikely that brands use refunds to react to sampling programs.

Table 4
Time series tests for S_1

Predictor variable: Price brand j	Criterion variable: Price brand i^a							Lag periods (m)
	1	2	3	4	5	6	7	
p_1	–	0.54	0.03	3.35	33.81	0.11	3.63	0
	–	15.89	10.92	7.49	40.93	11.08	13.78	4
	–	22.77	14.54	7.68	41.65	12.24	14.27	6
	–	23.35	19.58	7.93	43.56	14.31	15.79	8
	–	23.95	29.56	8.27	43.81	16.52	18.15	10
p_2	0.54	–	3.34	4.15	1.18	2.22	4.13	0
	3.62	–	7.32	4.80	13.21	3.94	6.56	4
	19.23	–	8.51	5.34	21.94	6.65	7.56	6
	34.84	–	13.22	7.59	40.86	8.40	19.68	8
	38.42	–	14.00	8.75	41.25	9.11	20.59	10
p_3	0.03	3.34	–	0.27	0.32	0.39	0.86	0
	3.45	8.73	–	4.99	4.57	4.51	11.19	4
	3.97	8.86	–	9.00	5.22	4.62	15.95	6
	4.57	9.00	–	11.30	5.33	5.34	18.47	8
	8.66	12.55	–	13.03	7.17	7.06	31.81	10
p_4	3.35	4.15	0.27	–	2.18	0.60	2.63	0
	4.23	7.08	5.94	–	2.83	2.09	2.83	4
	4.33	11.16	6.38	–	4.74	4.45	3.90	6
	5.78	11.46	7.14	–	6.04	6.04	6.45	8
	6.35	13.10	7.41	–	16.38	16.38	10.81	10
p_5	33.81	1.18	0.32	2.18	–	0.59	8.38	0
	34.88	13.30	5.51	3.53	–	12.27	21.60	4
	35.71	19.11	6.02	3.57	–	15.31	24.62	6
	40.15	22.06	7.70	6.75	–	22.77	24.91	8
	40.39	28.53	12.81	9.05	–	25.57	25.49	10
p_6	0.11	2.22	0.39	0.60	0.59	–	0.06	0
	11.25	11.31	4.05	4.52	4.39	–	2.54	4
	12.18	17.26	10.17	9.37	7.39	–	2.92	6
	12.65	23.84	12.20	10.42	8.01	–	5.44	8
	12.66	25.54	12.24	10.43	9.73	–	5.60	10
p_7	3.63	4.13	0.86	2.63	8.38	0.06	–	0
	6.88	7.47	6.24	3.10	12.93	3.51	–	4
	7.08	9.17	6.75	6.83	13.80	7.28	–	6
	9.26	13.28	7.59	10.62	14.38	14.24	–	8
	10.67	15.45	11.54	15.56	15.72	14.71	–	10

^a Bold figures indicate significant ($\alpha = 0.10$) causality.

Table 5

Causality test results ^a and statistically significant effects ^b in equations (6) and (7)

Criterion variable	Brand	Relevant predictors for each brand						
		1	2	3	4	5	6	7
Price	1		PF	F	PBF	P	P	PF
	2	PF*	F	P	PB	P	PR	P
	3	P	P	F		R	RB	
	4	P	PSB		SR		SBF	
	5	P	PBF	S	B	RF	F	PBF
	6	P			BF*	PR	RF	BF
	7	P	P	PS	BF	PS	RBF	
Sampling	2		F	SF	SF	F	SB	PS
	3		B		RBF	P	B	PSR
	4	F	SF	SF	PF		SBF	S
	5				PR	R		R
	6		SB	SF	SF	F	B	S
	7		B	S*F	F		BF	
Refund	4	F		P	PB	R		P*
	5	P	P	P	PR	P	PR	PBF
	6	P	PF	P	B	P	PF	PB
	7			S	PR	S	F	F
Bonus	2				PB	S		R
	4	PF	PB		R	PR	PR	PB
	6		B	PSF	F		F	PSR
	7	PF	P		B	S*R*	PR	RF
Feature	1	P		P	SF	PR	SRBF	SBF
	2	F	PS	F	S	PR*	RBF	S
	3	P	S	PS	S		SB	S
	4		B		B	P	BF	S*F
	5	F	B*	S	SF	P	S	PF
	6	F	SBF	F	S		PSR	S*RF
	7	P			PSBF	RF	P*SRF	RB

^a The causality test results are indicated by the presence of letters for brands that are determined to influence the criterion variable based on pairwise test statistics, using a maximum lag of 10 periods; P = price, S = sampling, R = refund, B = bonus, F = feature.

^b Bold letters indicate that the predictor variable has a statistically significant effect in the multiple regression as formulated in equations (6) and (7). If the sign of the coefficient for the predictor in the multiple regression is counter to expectations, the letter has the symbol * next to it.

Table 6

Number of significant causal relations

Predictor variable	Criterion variable					
	Price	Sampling	Refund	Bonus	Feature	Total
Price	24 ^a	4 (1) ^b	14 (3)	11	9 (5)	62 (9)
Sampling	5 (1)	14	2	4	18 (3)	43 (4)
Refund	5 (3)	4 (1)	4	6 (2)	7 (2)	26 (8)
Bonus	12	8 (1)	3 (1)	5	9 (2)	37 (4)
Feature	13 (4)	14 (2)	4 (2)	4 (2)	18	53 (10)
Total	59 (8)	44 (5)	27 (6)	30 (4)	61 (12)	221 (35)

^a The diagonal entries represent the number of simple competitive reactions.

^b The off-diagonal entries show the number of competitive reactions involving different variables, while the number in parentheses (not a subset of the number of competitive reactions) represents the number of relations between different variables for the same brand (intrafirm activities).

After the price–price relations, we see relatively high frequencies (18 in each case) for feature–feature and feature–sampling relations. Feature activity is perhaps the easiest, fastest, and least expensive way to react to competitive actions.

The numbers in parentheses indicate the frequency with which intrabrand relations are found. We note that the largest number of intrabrand relations for two different variables occurs with price and feature (5 for price–feature, and 4 for feature–price out of a maximum of 7). Interestingly, features account for the vast majority of all such intrabrand relations (22 out of 35). Features may, of course, be used to announce price cuts, refunds as well as bonus offers.

The causality test results shown in Table 5 are used to *specify* the dynamic-causal structure of the system of equations. That is, we use these statistical findings to provide indications of causal relations and to make equations (6) and (7) identifiable. The causality tests applied appear to give excellent indications of causal relations (see Leeflang, 1992).

Results

Equations (6) and (7) have been estimated using the original variables. Technically more appropriate steps were formulated by Haugh and Box (1977); see also Hanssens (1980). In all cases except one (see below), ordinary least squares was used to estimate the parameters. In the final equations, simultaneity does not exist, although we do have a number of recursive equations. The R^2 values vary from 0.48 to 0.95 and the DW values⁷ tend to be close to 2. Also, the R^2 values

range from 0.00 to 0.79 if competitive reactions with non-corresponding variables are excluded. Thus, there is substantial improvement in fit when multiple competitive reactions are allowed for.

We have identified all predictor variables with statistically significant effects in equations (6) and (7) using $\alpha = 0.05$ (two-tailed) by using bold letters representing the predictors in Table 5. For convenience of reporting, no information is provided on the specific lags for which significant effects obtained. However, the nature of lagged effects, broadly speaking, is summarized below.

For *price* the regression results include lagged endogenous variable coefficients with negative signs, consistent with alternating price movements. Generally the frequency with which competitive variables show up with significant coefficients is consistent with the results reported in Table 6. Thus, competing prices show up most frequently, followed by feature variables. Also, the coefficients generally have the expected signs.

Parallel price reactions to price changes are observed for brands 1, 4 and 5. Retailer-dominated short-run price reactions (to price changes) occur for brands 3, 5 and 6, and retailer-dominated long-run price reactions (to price changes) for brands 1, 5, 6 and 7. Finally, we observe manufacturer-dominated price reactions (to price changes) for brands 2, 3, 6 and 7. These distinctions are not shown in Table 5.

The causality tests reported earlier implied that brand 1 might be a price leader in this market. Consistent with this idea we find that brand 1's price changes show up as a significant predictor variable in one way or another for every one of the six competing brands' prices (see Table 5, column one). It is also of interest to note how many competing brands are represented with one or more predictor variables in each of the seven price equations. For example, for brands 5 and 7 all six competitors appear on the right-hand

⁷ Strictly speaking we have to use the Durbin statistic here. The values of the Durbin–Watson statistic only differ slightly from 2. Hence it is not necessary to compute the Durbin statistic to test for autocorrelation.

side, while for brand 4 only three competitors (brands 1, 2, and 6) show up. Thus, management of brand 4 may believe it is not as vulnerable to competitive actions.

The results also include information on *intrafirm* effects. For example, these effects exist for brands 3, 5 and 6 between the price and feature variables. The negative coefficients (not shown) confirm the belief that it is common to have a temporary price cut when a brand is featured. On the other hand, for brand 4 we have a positive coefficient (not shown) for the intrafirm effect associated with refunds. This may happen because management believes that refunds are attractive to consumers, and allow the firm to charge (temporarily) higher prices. Such a result is also consistent with the notion of price discrimination (see Narasimhan, 1984, and Vilcassim and Wittink, 1987).

There is little evidence of *sampling* reactions to price changes. For brands 4 and 6 the reactions occur to a large extent in response to competitive sampling (i.e., *simple* competitive reactions). Interestingly, for brands 3 and 5 the activity appears to be often in response to refund activity by competitors. Effects due to competitors' bonus offers show up for brands 3, 4, 6 and 7.

Intrafirm effects are obtained for brands 4 (with price), 5 (with refunds) and 6 (with bonus). In these cases the signs of the coefficients (not shown) are consistent with the idea that management may attempt to heighten the total impact of promotional programs by coordinating several variables. Thus, in *sales* response models one might expect a positive interaction effect for these variables as support for management's belief.

Results for *refund* reactions are obtained only for brands 4, 5, 6 and 7 (no refund activity exists for the other three brands). For three of the brands, reactions occur in response to refund activity by competing brands. However, all the other marketing variables show up at least once as having an

influence on refunds. Of particular interest is the large number of price variables showing up for brand 6. Its refund activity appears to be strongly influenced by price decreases for four competing brands. In addition, there is an intrafirm effect for price consistent with the idea of price discrimination.

For *bonus* reactions results are available only for brands 2, 4, 6 and 7. There is insufficient variability in bonus activity for the other three brands. For this variable evidence of simple competitive reactions is limited to brands 4 and 6. All other predictor variables have significant effects at least once, supporting again the need to accommodate multiple competitive reactions. In terms of the number of coefficients for competitive variables, we find the largest frequency for price followed by refunds, bonus, features, and sampling. Intrafirm effects occur for brand 7 only. For this brand, the refund coefficients suggest that bonus and refunds are substitutes.

The *feature* equations include lagged endogenous variables which for all brands, have a negative sign. Just as with the price variable, this implies alternating feature activities. The common joint use of price and feature variables may be responsible for this similarity.

Based on a sample count of the frequency with which significant coefficients show for competitive feature reactions (including multiple lagged effects which are not shown), we find the highest frequency for competitive featuring and sampling, followed by price, bonus offers and refunds. As expected, the sign for the competitive featuring variables is (always) positive.

Intrafirm effects are obtained six times. For brands 1 and 2 there is evidence of feature and price coordination. For brand 3 featuring and sampling may be substitutes. Similarly, for brand 7 features and refunds appear to be viewed as substitutes. On the other hand, management of brand 4 may

Table 7
Frequency of statistically significant effects for four types of competitive reactions

Criterion variable	Type of reaction ^a				
	Parallel movement	Short-run retailer dominated reaction	Long-run retailer dominated reaction	Manufacturer-dominated reaction	Total
Price	11	5	9	25	50
Sampling	3	1	13	20	37
Refund	4	0	7	6	17
Bonus	2	3	7	12	24
Feature	12	3	11	17	43
Total	32	12	47	80	171

^a We have excluded from this table a total of 11 statistically significant effects with the wrong sign, and all intrafirm effects; also, competitive reactions involving two or more lagged variables occurring for a given reaction type are counted as one effect. The total number of reactions in this table includes 39 effects more than can be identified in Table 5 (intrafirm effects excluded), because more than one type of reaction may occur for a given pair of variables.

believe that there are positive interaction effects between featuring and bonus offers.

With respect to the categorizations, stated in terms of parallel movements, short-run retailer-dominated reactions, long-run retailer-dominated reactions and manufacturer-dominated reactions, we provide a summary of the statistically significant effects in Table 7. For the five criterion variables we show the frequency with which each of the four types of reaction occur.

The last column in Table 7 shows the frequency distribution for the five marketing variables. This distribution differs from a uniform distribution ($p < 0.01$), and shows higher frequencies for price and feature, consistent with hypothesis H1. The last row in this table shows the total frequency with which each of the four reaction types occurs. This observed distribution also differs significantly ($p < 0.025$) from a uniform ⁸ pattern (i.e. the same frequency for the eleven periods, $t, t-1, \dots, t-10$). Reactions occur systematically more frequently the shorter the

lag, consistent with H3. This pattern differs marginally ($p < 0.15$) from the expected pattern if the type of reaction is independent of the criterion variable. The sample result suggests that featuring, in particular, in partial support of our hypothesis (H4), occurs relatively more frequently for shorter lags.

Discussion

In this paper we report on, we believe, the most extensive study of competitive reactions undertaken in marketing. Our results suggest that scanner data can be a useful basis for understanding the nature of competitive reactions implemented at the level of the retailer. The model used accommodates a variety of coordination and reaction functions, including parallel movements, retailer-dominated short- and long-run reactions, and manufacturer-dominated reactions.

Our findings show that *simple* competitive reaction functions fail to capture systematic effects due to other marketing variables. At the same time, simple reactions do account for a disproportionate number of reaction effects. Importantly, the estimated competitive reactions appear to be very complex.

⁸ The null hypothesis of a uniform pattern is tested by creating three categories: (a) t and $t-1$; (b) $t-2$ through $t-4$; (c) $t-5$ through $t-10$. The observed frequencies of 44, 47, and 80 are compared against expected frequencies under the null hypothesis of 31, 47 and 93.

The competitive reaction patterns observed can be used in a number of ways. Of central interest is its use in assessing the total effect of marketing activities employed for a brand. For example, instead of comparing the impact on brand sales (or market share or profits) for each of several alternative marketing instruments using only estimated *own* effects, one would add (subtract) the estimated *cross* effects multiplied by the reaction effects. Conclusions about the relative attractiveness of alternative marketing programs may change once these competitive reactions are incorporated. The complexity of the competitive reaction functions shown in this paper makes it especially likely that these conclusions change systematically and in perhaps unexpected ways.

There should also be some correspondence between the nature of competitive reactions and the effects the variables have on competing brands' sales. Everything else being equal, we expect firms to favor those instruments that can best maintain a brand in the position it may lose if a competitor's activity is not responded to. In the future, we intend to investigate how estimated competitive reaction patterns compare with marketing variables' estimated effects on competing brands' sales or market shares.

The findings can be used to infer management beliefs about degrees of competition between brands. For example, our results suggest that brand 1 is the price leader in this market. At the same time brand 1's price changes are stimulated by marketing activities for most of the competing brands. On the other hand, brand 4 appears to be reacting with price to activities for just three competitors and to the price of only brand 1.

The framework can also be used to quantify intrafirm marketing activities. For five brands we find an intrafirm effect for price and feature, consistent with the general belief that brands having a temporary price cut are likely to be featured. Evidence of coordi-

nation between other marketing variables occurs for brand 4 with respect to feature and bonus, and sampling and price. In all these cases, management apparently believes there are positive interactions in terms of sales effects for the variables involved.

Intrafirm patterns consistent with substitution are found for two brands. We have results that indicate a positive relation between refund activity and pricing for brands 4 and 6. The positive coefficients suggest that management may use a price increase to compensate for the cost of the refund activity. In addition, management may believe that consumers are less price sensitive when refunds are offered. And, to the extent that refunds are used by exposed consumers, price discrimination is indicated as well.

These results may provide useful information to (brand) managers about the extent to which competitors react to actions undertaken. However, these competitive reactions reflect only the activities actually carried out by retailers. Thus, our results represent what retailers implement, based at least partly on manufacturers' incentives.

We specified four general research hypotheses, all of which are found to be supported by the data. To the extent that one is willing to accept these hypotheses without empirical evidence, the consistency between hypotheses and data is a basis for claims about the validity of the approach used in this paper. Nevertheless, the research reported here is not without limitations. For example, the credibility of the results would be enhanced with evidence of predictive validity. We could not pursue this issue because our analysis required the entire sample. In this sense, our study may be considered exploratory.

It would also be useful to compare inferences about market structure based on the estimated competitive reactions with inferences based on alternative methods. An interesting question to pursue in future re-

search is whether managerial behavior (measured by competitive reactions) is less consistent over time than customer behavior. That is, consumers may show a higher degree of consistency in reactions to a given marketing program repeated over time for a given brand than a competing (brand) manager shows. It is of interest to explore this idea, and to consider the implications for competitive response models. Finally, aggregation issues (e.g. over stores, over varieties, over package sizes) and the separation of manufacturer and retailer roles deserve more research attention as well.

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